PARKINSON TREATMENT FRAMEWORK

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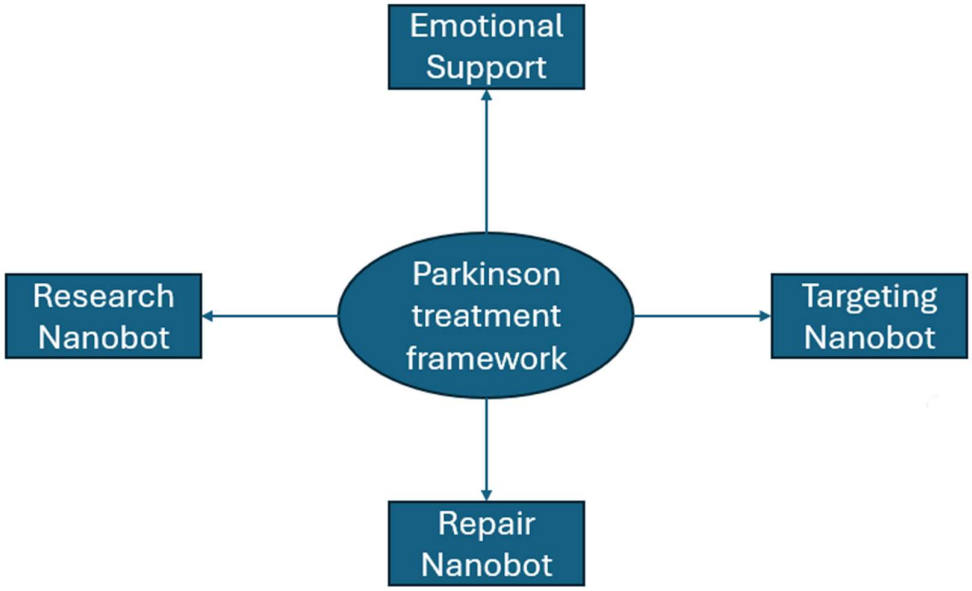
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Introduction

This Parkinson's treatment framework is designed to provide a reliable and safe solution for curing the disease. This framework was developed as Parkinson's is a major neurological disease that can cause extreme amounts of damage to a person's body and can even eventually indirectly lead to death. This framework focuses on the detection, treatment and emotional support in the aftermath.

This framework cannot promise a solution that is perfectly reliable, safe or efficient. Instead, this framework can promise to try and provide as reliable, safe and efficient solution as possible. This framework is for medical professionals and nanoengineers but anyone with interest is more than welcome to read this framework. It is important that the designs and systems used in this framework are adapted based on real-world results and standards.



The spider diagram above shows the main concepts that are involved within this treatment framework. We can see there are 3 types of nanobots instead of the usual 1 or 2 like in other treatment frameworks.

Parkinson's Detection Section

- **This Parkinson's detection framework has 5 steps.**
- **The first step is Parkinson's communication.**
- **This is where hospitals and medical professionals document and share Parkinson's symptoms online.**
- **This can be done by posting on social media, medical forums or communicating with friends and family.**
- **This step is important because patients need to know what may be wrong with them before visiting a medical professional.**
- **The second step is symptom detection**
- **This is where hospitals and medical professionals check the patient's current symptoms against known Parkinson's symptoms.**
- **This can be done by asking the patient questions, checking prior medical history and doing a check up on the patient.**
- **This step is important to ensure that the patient isn't facing another disease that may appear like Parkinson's.**
- **The third step is Parkinson's verification.**
- **This is where medical professionals check the patient for Parkinson's with more invasive and through examinations and tests.**
- **This can be done through tests such as Neurological exam, UPDRS, DaTscan, MRI, CT and PET scan.**
- **This step is important to ensure that the patient isn't facing other problems and to confirm the presence of Parkinson's.**
- **The fourth step is Parkinson's analysis.**
- **This is where medical professionals analyse the Parkinson's itself figuring out its extent and the stage it is in.**
- **This can be done through using tests like UPDRS, Hoehn and Yahr scale, MDS-UPDRS and CSF analysis**
- **This can also further be analysed through repeating the same tests from the verification stage consistently to track the progress.**
- **This step is important to figure out how bad the Parkinson's is and how urgent intervention may be needed.**
- **The fifth step is documentation.**
- **This is where doctors note down their analysis and the data they have on the patient in a structured and clear format**
- **This can be done by using documentation standards such as ICD, DSM-5, SNOMED CT and LOINC.**
- **It is important to remember that documentation can have any format as long as it's understandable to the doctor and their colleagues.**
- **This step is important because it allows doctors to remember key symptoms and act depending on what is happening.**

Parkinson's Nanobot Deployment Section

- **This nanobot deployment framework has 6 steps.**
- **The first step is nanobot training.**
- **This is where medical professionals and doctors train the nanobots to target Parkinson's specific cells.**
- **This can be done by providing the data for the nanobots, running simulations and performing testing with dead cells.**
- **This step is important as the nanobots need to know what they are targeting before they can be deployed within a body.**
- **The second step is nanobot verification.**
- **This is where medical professionals and doctors run diagnostics tests to ensure there are no issues with the nanobots.**
- **This can be done by using diagnostic tools on the nanobots, check connectivity between the nanobots and the external medical devices and checking the nanobot physically for any errors.**
- **This step is important to ensure that the nanobots have no errors before they enter the body.**
- **The third step is nanobot deployment**
- **This is where medical professionals deploy the nanobots into the human brain directly.**
- **This can be done through an Intrathecal which is the safest option for the patient while also allowing the nanobots to get to the brain.**
- **It is important that the targeting nanobots are deployed and used first before the repairing nanobot are deployed.**
- **This step is important because the nanobots need to bypass the BBB while not damaging the body.**
- **The fourth step is Nanobot functions.**
- **This is where the nanobots perform their actions as intended and as expected.**
- **This can be done by letting the targeting nanobots destroy the unhealthy cells and letting the repair bots repair damaged tissue.**
- **This step is important because it is the actual treatment that is used for Parkinson's**
- **The fifth step is nanobot transmission.**
- **This is where the nanobots transmit the data they have collected from destroying or repairing.**
- **This can be done by the nanobots sending out the signal with their data to the dedicated external medical device.**
- **The sixth step is nanobot disposal**
- **This is where the nanobots dissolve safely into a harmless form that the body can get rid of naturally.**
- **This can be done by giving the commands to dissolve or letting it do so by itself.**

Research Nanobot Design

- **This nanobot is designed to help researchers figure out how Parkinson's works and gather data on the cells that need to be targeted.**
- **I will begin by discussing the physical the components this nanobot will use.**
- **For size, this nanobot will be 4800 nanometres to account for all the components it needs to perform its jobs.**
- **For the power source, the nanobot will use a solid-state thin-film battery to be able to power itself for the logical functions.**
- **For the processing, the nanobot will use ASIC since it is one of the only available options for processing at nanoscale.**
- **For the storage, this nanobot will use RRAM since it is one of the best available options for nanoscale storage.**
- **For the transmitter, the nanobot will use Zinc Oxide (ZnO) thin-film piezoelectric material.**
- **For the sensors, the nanobot will use biodegradable electrochemical sensors, FET-Based Sensors, PH nano sensor.**
- **These sensors are needed to ensure that the researchers can collect as much data as possible on Parkinson and how the brain works.**
- **For cameras, the nanobot will use a carbon dot camera to ensure that the camera can fit at nanoscale.**
- **This camera is important as it gives researchers a visual view of what it looks like inside of the brain so they can see it.**
- **For the light source, the nanobot will also use carbon dots to provide light so the camera is able to see where it is going.**
- **This light source is important as without it the camera cannot see what's going on around it without light.**
- **For the movement, the nanobot will tiny but flexible mechanical legs to move around the brain.**
- **This is needed since the nanobot needs to move around.**
- **For electrical shielding, the nanobot will have a thin layer of magnesium around it to protect itself from the electrical impulses of the brain.**
- **I will now discuss the logical systems involved.**
- **For communication, the nanobot will need to provide real-time data to the external medical device.**
- **This real-time data will come from the nanobots multiple sensors and its cameras that it has.**
- **For navigation, the nanobot will need to use its mechanical legs to move throughout the brain.**
- **For disposal, the nanobot will ensure it has transmitted all data and will then dissolve itself.**
- **For safeguard, if the nanobot reaches 3% of battery or if the remote command is given then the nanobot will dissolve to prevent damage to the body.**

Targeting Nanobot Design

- **This nanobot is designed to target the cells that are causing Parkinson's once the research nanobots have provided the researchers with the information they need.**
- **I will begin by discussing the physical components.**
- **For size, the nanobot will be 4000-4500 nanometres**
- **For the power source, the nanobot will use the solid-state thin-film battery to ensure it is powered.**
- **This battery was chosen as it is an option that fits and it is the standard currently for my nanobots.**
- **For the processing, the nanobot will use the ASIC (Application-specific integrated circuit).**
- **This processing was chosen as it the smallest option while still providing reliable processing.**
- **For storage, the nanobot will use RRAM to store any data it collects from its sensors and destruction.**
- **For the transmitter, Zinc oxide (ZnO) thin-fil piezoelectric will be used as the material for that.**
- **This transmitter was chosen to be biocompatible and to use ultrasound that is done via the external medical device.**
- **For sensors, the nanobot will use biodegradable electrochemical sensors for detecting the cells that need to be dealt with.**
- **This was chosen as these sensors are biocompatible and are used as standard detectors for most my nanobots.**
- **For movement, the nanobot will use tiny but flexible mechanical legs that will allow it to move throughout the brain to find the cells.**
- **This was chosen as the electrical impulses of the brain would be too fast and uncontrollable for movement.**
- **I will now discuss the logical systems that this nanobot will need.**
- **For targeting, the nanobot will use Neprilysin and Cathepsin to bind and kill the alpha-synuclein.**
- **This was chosen as it will use the natural processes of the body instead of risking electrical shock damage.**
- **For navigation, the nanobot will use its sensors to detect the alpha-synuclein and use its mechanical legs to reach it.**
- **This was chosen since the nanobot needs to detect and move towards the target.**
- **For transmission, the nanobot will provide all its collected data to the external medical device to handle.**
- **For disposal, the nanobot will make to dissolve after its battery reaches 3% or the remote command is given.**

Repair Nanobot Design

- **This nanobot design is designed to let the body naturally repair the existing damage that Parkinson's has caused.**
- **I will begin by discussing the components that the nanobot will have physically.**
- **For size, the nanobot will be 4500 nanometres to meet the space needs of all the components especially the container.**
- **For the power source, the nanobot will use the solid-state thin-film battery to ensure it is powered.**
- **This battery was chosen as it is an option that fits and it is the standard currently for my nanobots.**
- **For the processing, the nanobot will use the ASIC (Application-specific integrated circuit).**
- **This processing was chosen as it the smallest option while still providing reliable processing.**
- **For storage, the nanobot will use RRAM to store any data it collects from its sensors and destruction.**
- **For the transmitter, Zinc oxide (ZnO) thin-fil piezoelectric will be used as the material for that.**
- **This transmitter was chosen to be biocompatible and to use ultrasound that is done via the external medical device.**
- **For sensors, the nanobot will use biodegradable electrochemical sensors for detecting the cells that need to be dealt with.**
- **This was chosen as these sensors are biocompatible and are used as standard detectors for most my nanobots.**
- **For movement, the nanobot will use tiny but flexible mechanical legs that will allow it to move throughout the brain to find the cells.**
- **This was chosen as the electrical impulses of the brain would be too fast and uncontrollable for movement.**
- **For the delivery mechanism, the nanobot will have a 100nm container to store about 8,000 BDNF.**
- **This was chosen as we can't afford to overstimulate the pain**
- **I will now discuss the logical systems needed for this nanobot**
- **For navigation, the nanobot will use its sensors to detect where the hippocampus is and use its legs to reach it.**
- **For transmission, the nanobot will use its transmitter to transfer all its collected data to the external medical device.**
- **For the disposal, the nanobot will dissolve once its BDNF has been released successfully, when it reaches 3% or if a remote command.**
- **For delivery, the nanobot will launch the BDNF once the nanobot detects that it has arrived at the hippocampus.**
- **This system was chosen as deploying BDNF anywhere else will fail.**

Parkinson's Aftermath Emotional Support Section

- **This emotional support framework has 5 steps, and it is needed as Parkinson's can cause mental health conditions such as depression.**
- **The first step is detection.**
- **This is where medical professionals examine the patient for symptoms that are afflicted with depression or other conditions.**
- **This can be done by talking to the patient, observing their behaviour and talking to any family or support system members.**
- **This step is important because mental health conditions can't be treated if they are unknown.**
- **The second step is identification**
- **This is where medical professionals look at the potential support or actions that need to be taken to support the patient.**
- **This can be done by looking at medical documentation of known treatment options and what can be realistically provided.**
- **This step is important because each mental health condition has its own way it needs to be treated.**
- **The third step is support offer.**
- **This is where medical professionals talk to the patient about support options and if they want to have support.**
- **This can be done by bringing the patient into a private room and presenting the options and ensuring that the patient doesn't feel forced.**
- **If a patient refuses support, they should not be forced to accept it as that can worsen that condition and make them feel uncomfortable.**
- **This step is important because patients need to feel like they are in control of the situation and the support they receive.**
- **The fourth step is recovery.**
- **This is where medical professionals implement the accepted support for the patient.**
- **This can be done by therapy sessions, using healthier coping mechanisms and having a strong support system.**
- **This step is important because the patient must be able to recover from the mental health condition to be fully cured of Parkinson's.**
- **The fifth and final step is monitoring**
- **This is where medical professionals keep an eye on a patient to ensure that there are no further medical issues.**
- **This can be done using appointments, observing patient behaviour and checking up on the patient's support system.**
- **This step is important because some patients can hide conditions as it is conditions like depression that do this.**
- **This emotional support framework can be adapted as needed.**

Parkinson's Treatment Patient Workflow

- **This patient workflow is designed to show the treatment process from the patient's viewpoint from start to finish.**
- **First, the patient sees a post about Parkinson's its symptoms and what to do if you see these symptoms**
- **Second, the patient remembers that and sees that their hands tremor when they are just resting.**
- **Third, the patient decides to book an appointment with their local general practitioner (GP).**
- **Fourth, GP asks the patient questions about what symptoms you have been experience**
- **Fifth, the GP determines that the patient may have Parkinson's or another disease that appears like it.**
- **Sixth, the patient is asked to book an appointment to see a neurologist to verify and eventually treat Parkinson's.**
- **Seventh, after a few days/weeks, the patient meets the neurologist during the appointment who begins to verify.**
- **The neurologist performs a Neurological exam, UPDRS, MRI and potentially a CT scan to verify the presence of Parkinson's.**
- **Eighth, the specialist confirms that the patient does have Parkinson's and asks for them to schedule an appointment for nanobot deployment.**
- **Ninth, a few days/weeks later, the patient comes in for the first nanobot deployment appointment.**
- **The research nanobots are deployed to the patient by an injection which might be painful for a second or two.**
- **Tenth, after the patient has waited for about 5 to 10 minutes, they are allowed to leave and asked to wait for a phone call to book another nanobot appointment.**
- **Eleventh, the patient arrives for their second nanobot deployment appointment.**
- **The targeting nanobots are deployed to the patient by the same injection which may still be patient for a second or two.**
- **Twelfth, the patient waited for 10 minutes maybe more, they are allowed to leave and asked to wait for a phone to book the final nanobot appointment.**
- **Thirteenth, the patient arrives for their third and final nanobot deployment appointment.**
- **The repair nanobots are deployed to the patient by the same injection which may still be patient for a second or two.**
- **Fourteenth, the patient is monitored for 1-2 weeks for side effects and offered emotional support during this time.**
- **Fifteenth, the neurologists confirm that Parkinson's is gone and that the patient can be discharged.**
- **The patient decides they don't need the emotional supports and goes home to live their life out before this disease came.**

Parkinson's Treatment Framework Trade-offs

- **This Parkinson's treatment framework has trade-offs just like any other system or framework I have created.**
- **I have identified 5 main trade-offs with this design**
- **The first trade-off is nanobot failure.**
- **This trade-off exists as nanobots aren't perfect meaning they can fail at any time during operation.**
- **This means that nanobots may not be able to complete their functions completely or even at all.**
- **This trade-off can be managed by deploying multiple nanobots for redundancy, monitoring progress and verifying nanobot functionality before deployment.**
- **The second trade-off is human error.**
- **This trade-off exists as humans aren't perfect meaning they can make mistakes during deployment, treatment and anywhere.**
- **This means that medical professionals may miss misconfigurations in the nanobots or fail to notice certain symptoms.**
- **This trade-off can be managed by automating processes where possible, having teams of professionals and ensuring workers have fair hours.**
- **The third trade-off is differentiation failure.**
- **This trade-off exists as stem cells will not always become new neurons as we need them to be.**
- **This is because within the brain the stem cells can become neurons, microglia and other cells like Astrocytes.**
- **This trade-off can be managed by having stem cells be produced from the brain and accepting that not every cell will become a neuron.**
- **The fourth trade-off is overstimulation.**
- **This trade-off exists as the repair nanobots will be delivering chemicals that encourage stem cell production and release within the brain.**
- **This means that if this delivered chemical is released in massive quantities it can cause uncontrollable amounts of stem cells that the body can't handle.**
- **This trade-off can be managed by controlling the amount of repair nanobots deployed, controlling the amount of BDNF and monitoring the patient for any signs.**
- **The fifth trade-off is complexity.**
- **This trade-off exists as it is difficult to understand and coordinate protocols for detection, nanobot deployment, emotional support and treatment.**
- **This means that staff may struggle to perform treatment for Parkinson's or any neurological disease that requires nanobots on this scale.**
- **This trade-off can be managed by training staff, making interfaces for devices clear and understandable and splitting the work across multiple professionals.**